

8

a Quadrant 3

b $\frac{\pi}{3}$ c $\frac{\pi}{3}$

9

a Quadrant 3

b $\frac{\pi}{7}$ c $\frac{\pi}{7}$

10

a 454° b -266°

c Quadrant 2

11

a 302° b -418°

c Quadrant 4

12

a 129° b 29°

13

a $\frac{\pi}{4}$ b $\frac{\pi}{3}$

14

a $360n$ b $360n$

15

a $2\pi n$ b $2\pi n$

Reference angles

16

a 69° b 17° c 85°

17

a i 193° ii 13° b i 159° ii 21° 18 30°



20 30°

21 a $\frac{14\pi}{13}$

b $-\frac{12\pi}{13}$

22 a $\frac{6\pi}{7}$

b $-\frac{8\pi}{7}$

23 a $\frac{21\pi}{11}$

b $-\frac{\pi}{11}$

24 $\frac{\pi}{11}$

25 $\frac{\pi}{7}$

26 $\frac{\pi}{5}$

27 $\frac{\pi}{3}$ radians

28 a $\frac{10\pi}{9}$ radians

b $\frac{\pi}{9}$ radians

29 a $\theta = \frac{\pi}{17}, \frac{16\pi}{17}$

b $\theta = \frac{18\pi}{17}, \frac{33\pi}{17}$

c $\theta = \frac{\pi}{17}, \frac{33\pi}{17}$

d $\theta = \frac{18\pi}{17}, \frac{16\pi}{17}$

e $\theta = \frac{\pi}{17}, \frac{18\pi}{17}$

f $\theta = \frac{16\pi}{17}, \frac{33\pi}{17}$

30 a $\frac{4\pi}{13}$

b i $\alpha = \frac{9\pi}{13}, \frac{4\pi}{13}$
iii $\alpha = \frac{9\pi}{13}, \frac{22\pi}{13}$

ii $\alpha = \frac{9\pi}{13}, \frac{17\pi}{13}$



31

a $\frac{\pi}{17}$

b

i $\alpha = \frac{18\pi}{17}, \frac{33\pi}{17}$

iii $\alpha = \frac{18\pi}{17}, \frac{\pi}{17}$

ii $\alpha = \frac{18\pi}{17}, \frac{16\pi}{17}$

32

a $\frac{2\pi}{17}$

b

i $\alpha = \frac{32\pi}{17}, \frac{19\pi}{17}$

iii $\alpha = \frac{32\pi}{17}, \frac{15\pi}{17}$

ii $\alpha = \frac{32\pi}{17}, \frac{2\pi}{17}$

33

a $\frac{8\pi}{17}$

b

i $\theta = \frac{25\pi}{17}, \frac{26\pi}{17}$

iii $\theta = \frac{8\pi}{17}, \frac{25\pi}{17}$

ii $\theta = \frac{8\pi}{17}, \frac{26\pi}{17}$

34

a $\frac{2\pi}{17}$

b

i $\theta = \frac{2\pi}{17}, \frac{15\pi}{17}$

iii $\theta = \frac{15\pi}{17}, \frac{32\pi}{17}$

ii $\theta = \frac{2\pi}{17}, \frac{32\pi}{17}$

35

a $\frac{5\pi}{13}$

b

i $\theta = \frac{5\pi}{13}, \frac{8\pi}{13}$

iii $\theta = \frac{5\pi}{13}, \frac{18\pi}{13}$

ii $\theta = \frac{8\pi}{13}, \frac{18\pi}{13}$